INCH-POUND
MIL-DTL-16377/82A
w/AMENDMENT 1
25 July 2022

SUPERSEDING
MIL-DTL-16377/82A
30 June 2014

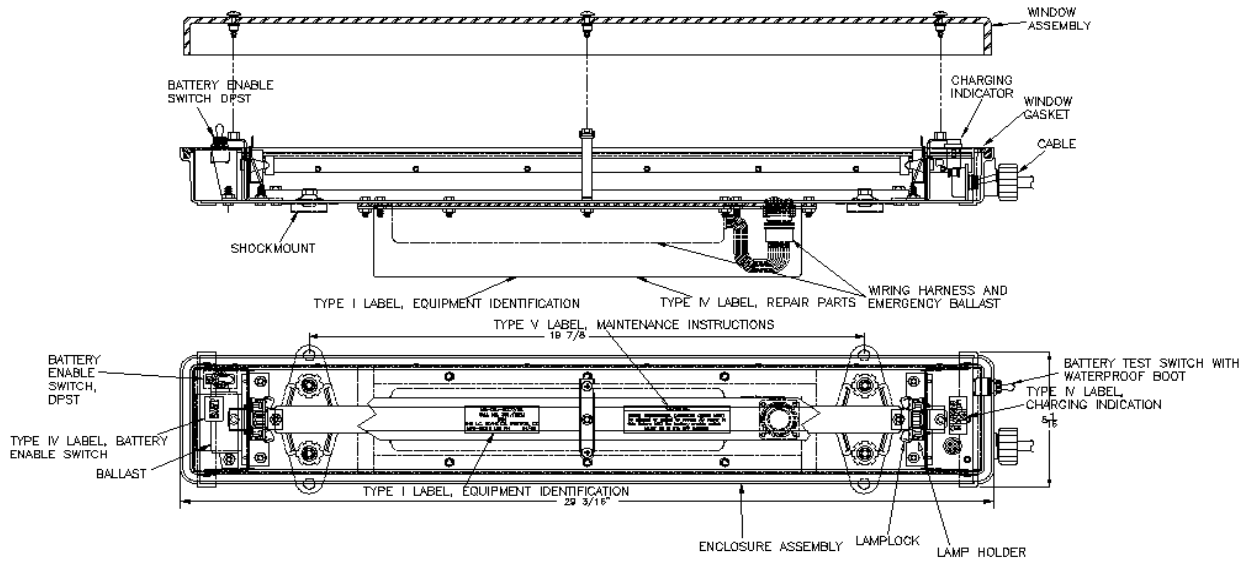
DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; T8 FLUORESCENT, EMERGENCY LIGHTING, 17 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 331.1T8EM, 331.2T8EM, 347.2T8EM, AND 347.3T8EM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377. If specified by the acquisition activity, solid state lighting (SSL) fixtures designed and manufactured in accordance with MIL-DTL-16377/73 may be substituted for fixtures manufactured in accordance with this specification sheet.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.



NOTE:

1. Symbols 331.2T8EM and 347.3T8EM use identical construction with flush mounting flange not shown.

FIGURE 1. Typical view of symbols 331.1T8EM and 347.2T8EM.

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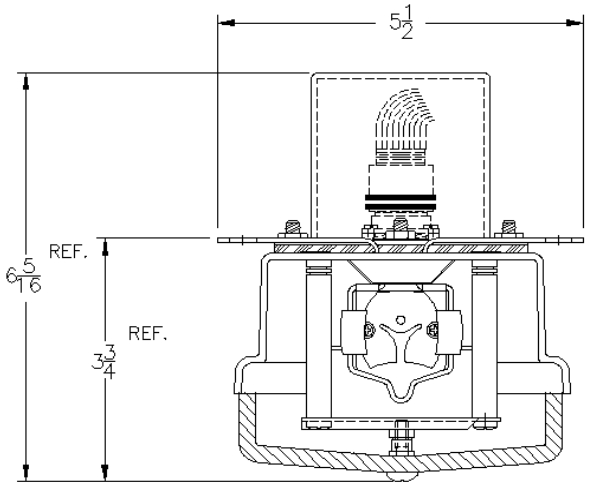
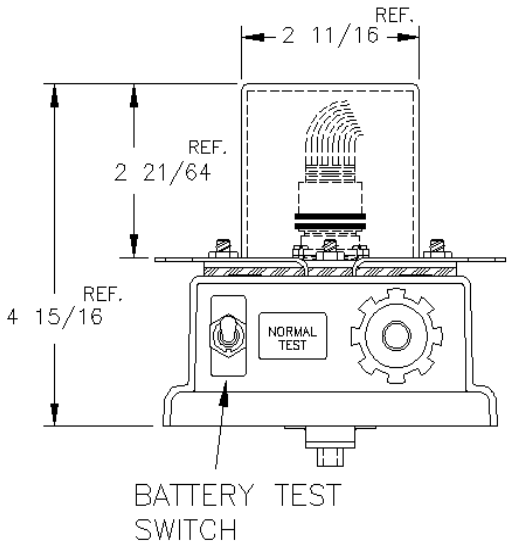


FIGURE 2. Typical section for symbols 331.1T8EM and 347.2T8EM.



NOTES:

1. Symbols 331.2T8EM and 347.3T8EM use identical battery test switch construction with additional switch hardware (not shown) for switch operation below ceiling.
2. Type II label required on all fixtures.

FIGURE 3. Typical section for symbols 331.1T8EM and 347.2T8EM.

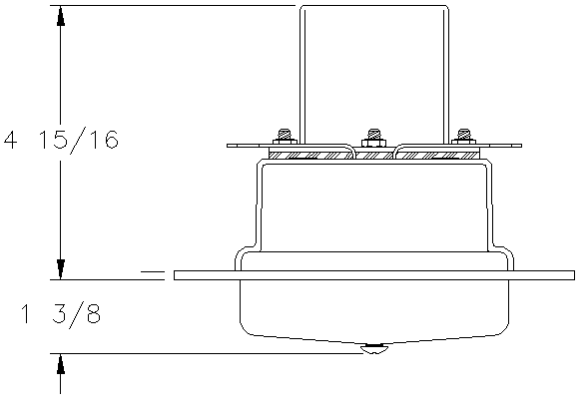
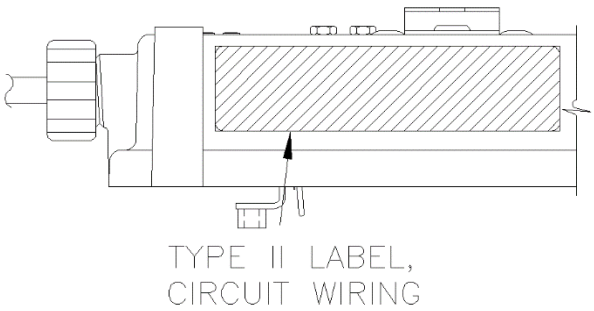


FIGURE 4. Typical section for symbols 331.2T8EM and 347.3T8EM.



NOTE:

1. Type II label required for all fixtures.

FIGURE 5. Typical side view for symbols 331.1T8EM, 331.2T8EM, 347.2T8, and 347.3T8EM.

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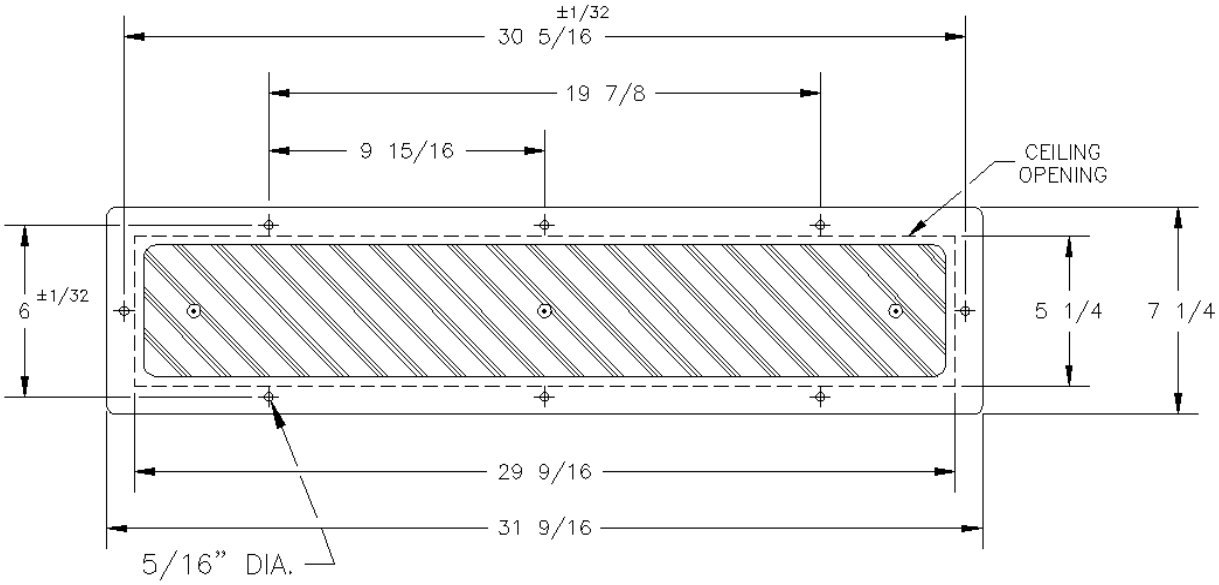


FIGURE 6. Flush mounting dimensions for symbols 331.2T8EM and 347.3T8EM.

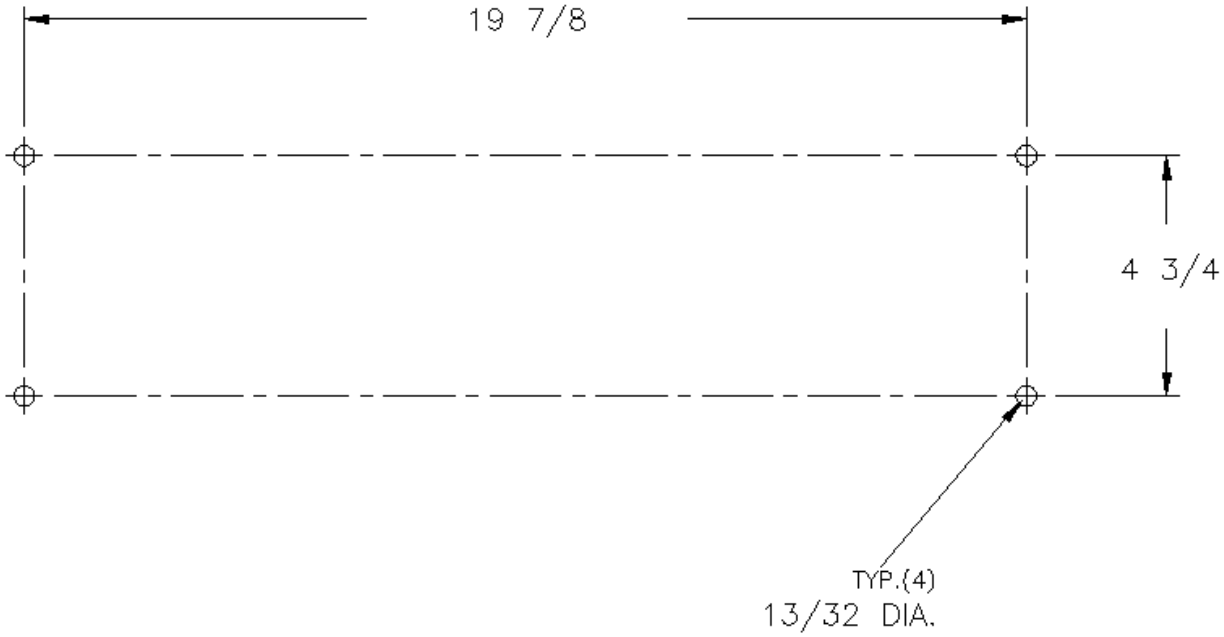
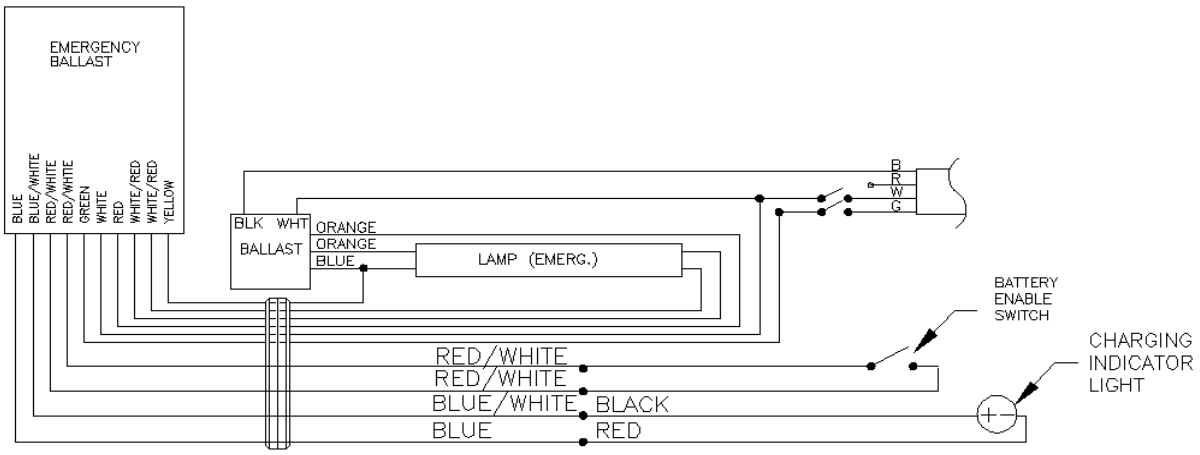


FIGURE 7. Standard mounting dimensions utilizing shock mounts for symbols 331.1T8EM and 347.2T8EM.

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NOTES:

1. The black input conductor is the only switched power lead to the fixture, used for normal external on/off lamp control.
2. The white and green input conductors shall be connected directly to the power source and remain continuously energized for emergency lamp battery charging. The white and green input conductors shall not have power interrupted by on/off control switches.
3. The white input conductor is shared by the normal lamp circuit and by the emergency circuit.

FIGURE 8. Typical wiring diagram.

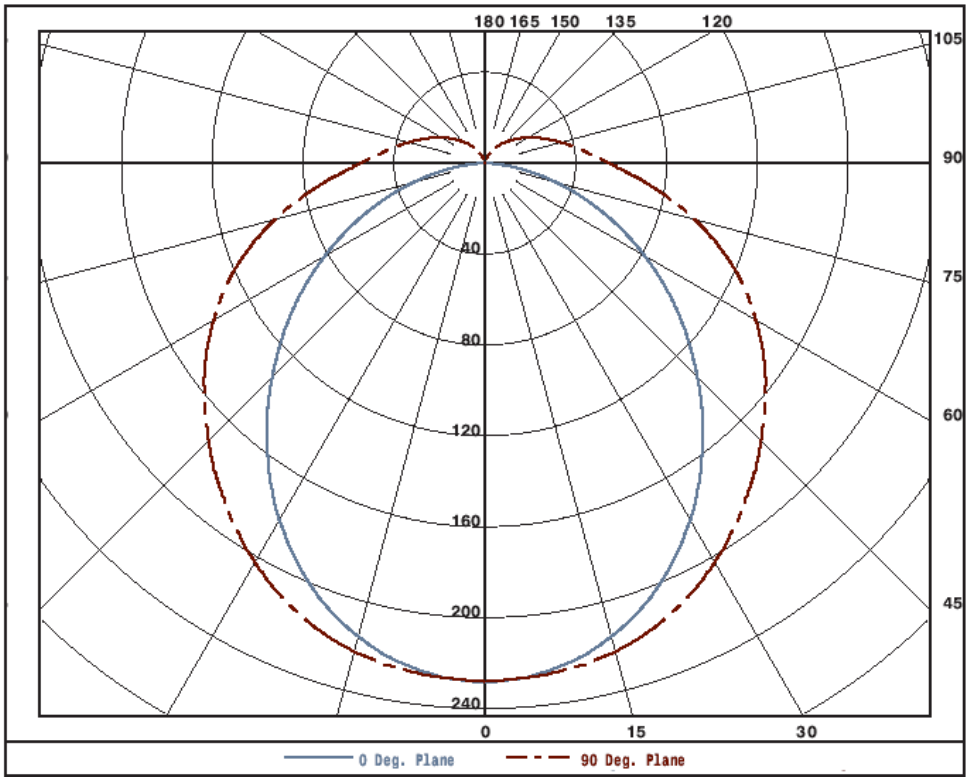


FIGURE 9. Candela distribution curves, white diffusing window, symbols 331.1T8EM and 331.2T8EM.

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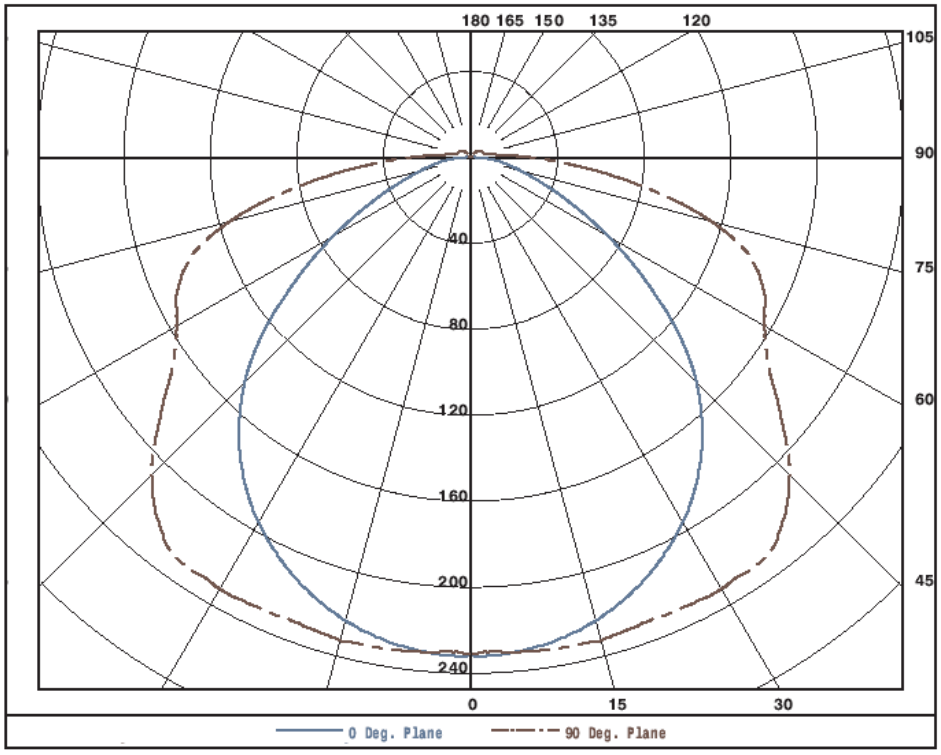


FIGURE 10. Candela distribution curves, prismatic window, symbols 347.2T8EM and 347.3T8EM.

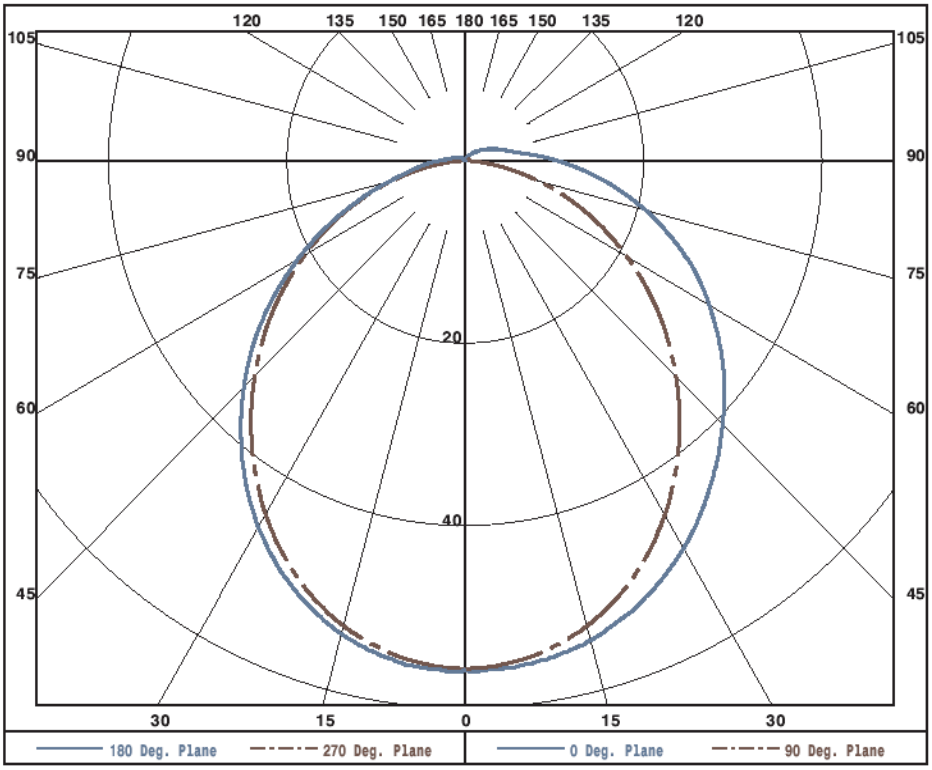


FIGURE 11. Candela distribution curves, emergency mode.

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TABLE I. Numerical candela distribution, white diffusing window, symbols 331.1T8EM and 331.2T8EM.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	228	228
5	227	227
15	216	223
25	195	211
35	166	193
45	132	172
55	98	150
65	64	125
75	32	95
85	7	66
90	2	55

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	1	46
105	0	35
115	0	26
125	0	19
135	0	13
145	0	8
155	0	4
165	0	0
175	0	0
180	0	0

TABLE II. Numerical candela distribution, prismatic window, symbols 347.2T8 and 347.3T8.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	231	231
5	231	231
15	223	233
25	207	231
35	182	231
45	147	208
55	97	167
65	52	149
75	23	115
85	9	51
90	2	25

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	2	13
105	2	6
115	1	6
125	0	5
135	0	4
145	0	2
155	0	1
165	0	0
175	0	0
180	0	0

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TABLE III. Numerical candela distribution, emergency mode.

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
0	56	56
5	56	55
15	54	53
25	51	48
35	46	41
45	41	33
55	35	25
65	28	17
75	21	8
85	14	2
90	10	1

Vertical Angle	Candela Distribution	
	0-Degree	90-Degree
95	8	0
105	4	0
115	3	0
125	2	0
135	1	0
145	1	0
155	0	0
165	0	0
175	0	0
180	0	0

PHOTOMETRIC DATA:

Photometry reports based on LM-41 and other pertinent Illuminating Engineering Society (IES) procedures.

Plane:	0 degrees is parallel to lamp, 90 degrees is perpendicular to lamp.
Electrical values:	120 volts, 60 hertz, 18.0 watts + 4.0 watts (battery charging circuit) = 22.0 watts total.
Ambient temperature:	77±1.8 °F (25±1 °C).
Lamps:	GE F17T8-SP41-ECO, or equal, (seasoned for at least 100 hours).
Lamps correction:	Lamps rated at 1,325 lumens.
Ballast:	Furnished with fixture, see MIL-DTL-16377/81 for 17-watt lamp specification and MIL-DTL-16377/77 for emergency ballast specification.

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REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBERS:

331.1T8EM:	Complete lighting fixture assembly, standard mounting with one 17-watt T8 cool white fluorescent lamp and white diffusing window assembly; includes emergency battery/ballast circuit for emergency lighting.
331.2T8EM:	Complete lighting fixture assembly, flush mounting with one 17-watt T8 cool white fluorescent lamp and white diffusing window assembly; includes emergency battery/ballast circuit for emergency lighting.
347.2T8EM:	Complete lighting fixture assembly, standard mounting with one 17-watt T8 cool white fluorescent lamp, clear and white prismatic window assembly; includes emergency battery/ballast circuit for emergency lighting.
347.3T8EM:	Complete lighting fixture assembly, flush mounting with one 17-watt T8 cool white fluorescent lamp, clear and white prismatic window assembly; includes emergency battery/ballast circuit for emergency lighting.

DESIGN:

Electrical rating:	17 watts, 120 volts, 60 hertz.
Dimension and configurations:	See figures 1, 2, 3, and 4.
Photometric data:	See figures 9, 10, and 11 and tables I, II, and III.
Wiring diagram:	See figure 8 for typical wiring.
Ballast:	See MIL-DTL-16377/81 for 17-watt T8 lamp specification and MIL-DTL-16377/77 for emergency ballast specification.
Mounting dimensions:	
Symbols 331.1T8 and 347.2T8:	Standard mounting (surface), see figure 7.
Symbols 331.2T8 and 347.3T8:	Flush mounting (recessed), see figure 6.
Window assembly:	
Symbols 331.1T8 and 331.2T8:	White window, see table IV and figure 27 of MIL-DTL-16377.
Symbols 347.2T8 and 347.3T8:	Prismatic window, see table IV and figure 27 of MIL-DTL-16377.
Enclosure:	Watertight, see 3.5.11 of MIL-DTL-16377.
Cable:	10 feet (minimum) of Type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 331.1T8EM and 347.2T8EM:	Not to exceed 13.5 pounds.
Symbols 331.2T8EM and 347.3T8EM:	Not to exceed 15 pounds.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 1 and 5:

a. Two Type I labels, equipment identification, reference MIL-DTL-16377 figure 11; shall include manufacturer and address, one located on back and other located on an inner nonreflecting surface.

b. One Type II label, circuit wiring information, reference MIL-DTL-16377 figure 12; located on outside surface adjacent to cable entrance.

c. Two Type IV labels, repair parts, reference MIL-DTL-16377 figure 14; information to be shown shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by a list of all standard parts shown in table IV, located on an inner nonreflecting surface and back of fixture.

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d. One Type IV label, reference MIL-DTL-16377 figure 14; information to be shown shall be "CHARGING INDICATOR LIGHT."

e. One Type IV label, reference MIL-DTL-16377 figure 14; information to be shown shall be "BATTERY ENABLE."

f. One Type V label, reference MIL-DTL-16377 figure 15; information to be shown shall be "DURING MAINTENANCE, PROTECTIVE DEVICE MUST BE OPENED OR PULLED TO SECURE AC POWER TO THIS FIXTURE AND THE BATTERY ENABLE SWITCH MUST BE IN THE OFF POSITION."

EXAMINATIONS AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article testing/inspection:
Complete fixtures: See table V.
Window assemblies: See table VI.
Conformance:
Comparison inspection: See tables V and VI.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is show in table VII.

NOTES:

1. Installation circuit wiring shall incorporate a minimum ratio of eight 17-watt lamps and one emergency 17-watt lamp. The 8-to-1 ratio is required for meeting MIL-STD-1399-300-1 harmonic requirements. The 17-watt emergency ballast produces non-linear current that shall be offset by the linear current from eight 17-watt lamp ballasts connected to the same circuit.
2. Emergency fixtures may be designated as Navy Emergency Automatic Lighting System (NEALS).
3. The "battery enable" switch shall be placed in the normal position after installation.

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TABLE IV. Standard parts.

Name	Military P/N or Specification	National Stock Number (NSN)	Remarks
Lamp	Commercial identification F17T8/CW	6240-01-391-6027	
Lamplock	M16377/40-003		
Lampholder	UL 542	6250-00-284-0481	Commercial
Ballast	M16377/81-001		Integrated starter type
High-temperature ballast	M16377/81-002		High-temperature applications
High-temperature starter	FS-4	6250-00-179-2354	High-temperature applications
Startersocket	UL 496	6250-00-295-2738	High-temperature applications
Shockmount	M16377/41-008	6210-01-393-4302	0.067 steel, zinc plated
Window assembly, diffusing	M16377/8-001	6210-00-548-0601	1/
Window assembly, prismatic	M16377/8-003		1/
Emergency ballast, wiring harnesses, and mating connector	M16377/77-001	6250-01-491-7039	Complete assembly: includes wiring harnesses and mating AMP #206043 connectors.
Charging indicator light		6220-01-415-7889	LED type
Switch, battery enabled		5930-01-414-9240	DPST
Switch, battery test	Carling 2167-M-A-NC-73XCA4 or equal		Momentary toggle
Switch cover, rubber boot	MIL-DTL-5423/2		Neoprene
Filter, 17-watt T8 fluorescent lamp (Dark adaptation)	M16377/42-010	6210-01-313-8169	Red transparent
Filter, 17-watt T8 fluorescent lamp (1% transmission gray)	M16377/42-011		
Filter, 17-watt T8 fluorescent lamp (2.5% transmission gray)	M16377/42-012		
Filter, 17-watt T8 fluorescent lamp (Amber)	M16377/42-015		
Filter, 17-watt T8 fluorescent lamp (Broad band blue)	M16377/42-016		

NOTE:

1/ Window assemblies (diffusing and prismatic) are the same as T12 lamp fixtures (MIL-DTL-16377/8).

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TABLE V. First article tests and comparison inspection for complete fixtures. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Electromagnetic interference	3.6.14	4.8.16	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	2/
Voltage transient spike	3.6.18.2	4.8.23	3/
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Enclosure effectiveness	3.5.11	4.8.14.4	4/ Watertight
Shock	3.5.12	4.8.7	4/
Vibration	3.5.13	4.8.8	4/
Enclosure effectiveness	3.5.11	4.8.14.4	4/ Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	4/
Insulation resistance	3.6.16	4.8.3	4/
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Light output	3.7.1	4.8.6.1	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	
Emergency light output: Distribution curves shall be documented during emergency lamp operation	3.7.1	4.8.6.1	5/
Emergency ballast performance: Operating time and discharge test	See MIL-DTL-16377/77, table II, battery discharge test	See MIL-DTL-16377/77, table II, battery discharge test	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ If less, then 1 KVA exception not applicable for input waveform total harmonic distortion.

3/ Solid state components shall be subject to 250 positive spikes and 250 negative spikes (500 total spikes), one spike per minute, using testing method identified in MIL-STD-1399-300-1.

4/ Tests shall be performed in the order specified.

5/ Shall meet or exceed published distribution curves.

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TABLE VI. First article tests and comparison inspection for window assemblies. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/ Watertight
Light output	3.7.1	4.8.6.1	Candela distribution

NOTES:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.
2/ Tests shall be performed in the order specified.

TABLE VII. Complete fixture part numbers.

Military Part Number M16377/82-	Item Name	National Stock Number (NSN)
331.1T8EM	Complete fixture, Symbol 331.1T8EM	6210-01-595-4129
331.2T8EM	Complete fixture, Symbol 331.2T8EM	
347.2T8EM	Complete fixture, Symbol 347.2T8EM	
347.3T8EM	Complete fixture, Symbol 347.3T8EM	

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Army – CR4
Navy – SH
Air Force – 03

Review activities:
Navy – AS, CG
DLA – GS2

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
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